ISOSORBIDE DINITRATE

Never with PDE-5 inhibitors

Nitrate-free interval

CHEMICAL NAME 1,4:3,6-dianhydro-D-glucitol 2,5-dinitrate
CLASS Organic nitrate (antianginal vasodilator)
TRADE NAMES Isordil · Sorbitrate · Isoket (IV/spray) · BiDil (with hydralazine)

THERAPEUTIC USES / INDICATIONS

- Prophylaxis of angina.
- Acute angina (sublingual).
- Heart failure (with hydralazine).
- Acute left ventricular failure (IV).

ADMINISTRATION

- **PO** tablets 2–3 times daily (asymmetric dosing e.g. 8 am, 2 pm) to allow nitrate-free interval.
- **SL** tablets for acute attacks.
- **IV** infusion via syringe pump.

DOSAGES

- **Angina prophylaxis:** 10–20 mg three or four times daily (UK); allow a daily nitrate-free interval (≥14 h for IR, US label).
- **Acute angina SL:** 5–10 mg.
- **Heart failure (with hydralazine, e.g. BiDil 20/37.5 mg):** 20 mg three times daily, titrated to 40 mg three times daily.
- **IV:** usually 2–12 mg/h (up to 20 mg/h); avoid PVC/PU giving sets.

PHARMACOKINETICS

- **Onset:** SL 2–5 min; PO 15–40 min. **Duration:** SL 1–2 h; PO 4–6 h.
- Extensive first-pass metabolism to active isosorbide mononitrate.
- Excreted in urine. **t½:** 1 h (parent).

PHARMACODYNAMICS

- NO donor → ↑ cGMP → venous and arterial dilation; ↓ preload and myocardial oxygen demand.
- Coronary vasodilation.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, flushing.
- Dizziness, postural hypotension.
- Tachycardia.

ADVERSE EFFECTS (SERIOUS)

- Severe hypotension.
- Tolerance.
- Methaemoglobinaemia (rare).

CONTRAINDICATIONS & CAUTIONS

- PDE-5 inhibitor use.
- Hypotension, hypovolaemia.
- Severe aortic stenosis, HOCM, constrictive pericarditis.
- Angle-closure glaucoma (caution).

KEY INTERACTIONS

- **Sildenafil, tadalafil:** severe hypotension.
- **Antihypertensives, alcohol:** additive.

NURSING MANAGEMENT

- Check BP before dose.
- Ensure nitrate-free interval in dosing times.
- Monitor angina frequency and headache.
- Fall precautions.

PATIENT EDUCATION

- Take at the prescribed times (not at bedtime) to prevent tolerance.
- Headache usually eases after a few days; paracetamol helps.
- Never take with erectile dysfunction drugs.
- Rise slowly.

KEY POINTS

- Nitrate for angina prophylaxis.
- Asymmetric dosing — nitrate-free period.
- Metabolised to mononitrate.
- No PDE-5 inhibitors.

SOURCES CHECKED **[356]** Isosorbide Dinitrate 0.05% w/v Solution for Injection or Infusion (UK SmPC) **[357]** Isosorbide Dinitrate, oral (US label) **[358]** Isosorbide Dinitrate, oral (US label, Bryant Ranch Prepack) **[348]** BiDil, oral (US label)

Full references p. 432–457